

# When Fairness Metrics Disagree Under Shift: A Stylized Case Study of a Medical Follow-up Risk Score under Hospital Deployment Shift

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July 24, 2026

## 1 Introduction

Group fairness metrics are typically defined under a fixed data-generating process, yet real-world deployments—such as moving a medical follow-up risk score across hospitals—involve distribution shifts that can make these metrics behave unpredictably. In this project, we ask: (1) how do standard group fairness metrics behave under controlled shifts even when the underlying classifier remains fixed? (2) under what conditions can demographic parity remain stable or appear favorable while error-based metrics degrade? and (3) how do importance reweighting, post-hoc threshold adjustment, and target-distribution retraining affect fairness distortion and what trade-offs do they incur? We address these questions via a synthetic simulation framework mimicking a cross-hospital deployment, which allows us to systematically sweep over shift types and magnitudes.

## 2 Related Work

Fairness under non-stationary environments has been studied in several recent works. Asiaee and Aryan (2026) examined invariance under group-conditional prior shifts, while Chen et al. (2022) studied fairness transferability under bounded shifts. Barrainkua et al. (2022) and Shao et al. (2024) provided comprehensive surveys on preserving fairness guarantees and supervised fairness under distribution shifts. However, existing work rarely dissects, under a unified medical deployment simulation, how a static demographic parity value can systematically mask the degradation of error-based metrics—the central gap our study addresses.

## 3 Method

### 3.1 Distribution shift mechanisms

We consider three families of controlled distribution shift implemented via `generate_data`: (i) group-conditioned base-rate shift, (ii) group-specific covariate shift, and (iii) group-specific label noise shift. The source classifier is held fixed for the diagnostic baseline, while mitigation experiments additionally evaluate reweighted training, post-hoc thresholding, and target-distribution retraining. Full mathematical definitions of each shift mechanism are deferred to Appendix A.

### 3.2 Fairness and calibration metrics

We report standard fairness quantities via `fairlearn.metrics`: demographic parity difference (absolute difference in positive prediction rates), equalized odds difference (worst-case TPR/FPR discrepancy), and the true-positive-rate gap (equal-opportunity violation). We also track global and group-wise expected calibration error (ECE) and the ECE gap to capture calibration disparities. Formal definitions of all metrics are provided in Appendix B.

### 3.3 Mitigation strategies

We evaluate three interventions: (i) KDE-based importance reweighting using an unlabeled target adaptation sample, (ii) post-hoc threshold adjustment targeting equal opportunity on a labeled calibration set, and (iii) target-distribution retraining on an independent shifted sample. Detailed implementation of each strategy is given in Appendix C.

## 4 Experimental Setup

### 4.1 Shift parameter grid

We sweep over the following grids:

- Group-conditioned base-rate shift:  $\alpha \in [0, 1]$  with 11 steps.
- Covariate shift:  $\gamma \in [1, 4]$  with 10 steps.
- Asymmetric label noise:  $\beta \in [0, 0.3]$  with 10 steps.

Implementation details (sample sizes, classifier hyperparameters, ECE binning, and random seed) are reported in Appendix D.

## 5 Results

### 5.1 Key empirical findings

Across all three shift families, we observe that demographic parity, error-based metrics, and calibration can move in markedly different directions. Our main results are summarized below.

#### 5.1.1 Diagnosis (RQ1 & RQ2):

- Under **group-conditioned base-rate shift**, DP increases from 0.01 to 0.26 as  $\alpha$  goes from 0 to 1, while the TPR gap remains between approximately 0.03 and 0.06. The two metric families therefore react very differently to the same prevalence change.
- Under **covariate shift**, DP grows from 0.01 to 0.18, EO reaches 0.30, and the TPR gap reaches 0.12 at  $\gamma = 4$ . The ECE gap also becomes large, reaching approximately 0.32. In this regime several metrics worsen together, but at different rates, illustrating that they need not signal the same level of concern.
- Under **asymmetric label noise**, DP stays artificially flat at 0.01 even as the TPR gap worsens from 0.03 to 0.18. This is the most deceptive regime from a clinical equity standpoint: a static DP value would suggest that the referral rate is balanced across groups, yet the underlying TPR gap implies that one group is being systematically under-referred for necessary follow-up.

### 5.1.2 Mitigation (RQ3):

- **KDE Reweighting:** Nearly ineffective. The ESS fraction ranges from 0.762 to 0.950 across the experiments, so clipping does not collapse the effective sample, but the resulting DP and EO curves remain close to the frozen baseline. This suggests that density-ratio reweighting alone is insufficient to recover fairness in this setting, especially when the shift affects label prevalence or label noise rather than only the marginal feature distribution.
- **Threshold Adjustment:** Under covariate shift it reduces the maximum TPR gap from 0.124 to 0.018, yet the maximum EO difference rises from 0.298 to 0.408 because the FPR gap grows. Under label shift it lowers the maximum TPR gap from 0.176 to 0.030 but raises DP from approximately 0.012 to as much as 0.178; at  $\beta = 0.3$ , the calibrated threshold for group B is 0.16.
- **Target Retraining:** Under severe covariate shift ( $\gamma = 4$ ), balanced accuracy falls from 69.8% for the frozen source model to 53.7% after target retraining. This indicates a limitation of the present logistic model and ERM objective under the shifted distribution, rather than a universal separability limit. Under label noise, retraining keeps DP low but only modestly changes EO.

### 5.2 Phase diagrams of metric distortion

To highlight non-monotonic and counterintuitive behavior, we construct “phase diagrams” where each point corresponds to a particular shift configuration and is colored by the value of a fairness metric.

We emphasize regions where demographic parity remains stable or changes slowly while error-based metrics such as EO and the TPR gap worsen. The mitigation plots also show the converse: explicitly reducing the TPR gap can increase DP or the FPR-driven EO difference. Thus, a favorable movement in one fairness quantity need not correspond to uniformly fairer behavior. Diagrams are presented in Appendix E.

## 6 Conclusion and Future Work

Our simulations reveal several patterns. A stable or comparatively favorable demographic-parity value can mask large changes in error-based fairness; the asymmetric-label-noise experiment is the clearest example. In practice, this means that a hospital dashboard that only monitors demographic parity may fail to detect inequities in access to follow-up care. In our stylized setting, this would naturally lead a hospital deploying a follow-up risk score to track at least DP, TPR gaps, and calibration gaps side by side. Error-based metrics such as equalized odds can degrade sharply under certain shifts, even when the classifier is unchanged. This suggests that error-based fairness metrics should be monitored alongside DP in deployment, especially when the target population differs from the training population. In this setting, the TPR gap is especially relevant because missed necessary follow-up can correspond to unequal access to care. Different metrics can therefore disagree not only in level but in trend as the environment drifts. For clinical decision support systems, this implies that fairness monitoring cannot rely on a single metric; the choice of metric should be guided by the specific equity concern (e.g., equal access vs. equal accuracy).

We presented a synthetic simulation framework showing that stable demographic-parity values can mask degradation in error-based fairness, and that different fairness metrics can disagree in both level and trend under distribution shifts. These findings caution against over-relying on any single statistic as a stable guarantee when deploying models across hospitals. Future work includes extending the framework to more realistic feature spaces and multi-class settings, repeating sweeps over multiple random seeds with uncertainty bands, comparing richer model families and randomized equalized-odds post-processing, and exploring causal and counterfactual perspectives on fairness under non-stationarity.

## References

- [1] Asiaee, A., & Aryan, K. (2026). Fairness under group-conditional prior probability shift: Invariance, drift, and target-aware post-processing. arXiv preprint arXiv:2602.05144.
- [2] Chen, Y., Raab, R., Wang, J., & Liu, Y. (2022). Fairness transferability subject to bounded distribution shift. arXiv preprint arXiv:2206.00129.
- [3] Barrainkua, A., Gordaliza, P., Lozano, J. A., & Quadrianto, N. (2022). A survey on preserving fairness guarantees in changing environments. arXiv preprint arXiv:2211.07530.
- [4] Shao, M., Li, D., Zhao, C., Wu, X., Lin, Y., & Tian, Q. (2024). Supervised algorithmic fairness in distribution shifts: A survey. arXiv preprint arXiv:2402.01327.

## A Full Distribution Shift Mechanisms

This appendix provides the complete mathematical definitions for the three shift families summarized in Section 2.1 of the main text. Each is implemented as a wrapper around the base data generator `generate_data` (`src/shifts.py`).

### A.1 Group-conditioned base-rate shift

The function `group_shift(severity)` varies the group-conditioned base rates of the positive class while keeping the feature generator otherwise fixed. Concretely, we define

$$\begin{aligned}\Pr(Y = 1 \mid S = 0) &= 0.3 + 0.2 \cdot \alpha, \\ \Pr(Y = 1 \mid S = 1) &= 0.3 - 0.2 \cdot \alpha,\end{aligned}$$

with  $\alpha \in [0, 1]$ . Increasing the severity therefore simultaneously increases the positive rate for group A and decreases it for group B. The function `group_shift` passes the corresponding base rates to `generate_data` and returns the resulting features  $X$ , labels  $Y$ , and group labels  $S$ .

This family of shifts induces asymmetric changes in label prevalence across groups while keeping the conditional feature generator  $P(X \mid Y, S)$  fixed. Because  $X$  depends on  $Y$ , its marginal distribution can still change as the base rates change, even though the conditional generator is fixed; from the perspective of an observer who does not see  $Y$ , this may resemble a covariate shift. In a hospital deployment context, this corresponds to a scenario where the target hospital serves a patient population with a different group-specific prevalence of the underlying condition (e.g., different referral patterns or catchment demographics), while the clinical manifestation of the condition given group membership remains the same.

### A.2 Group-specific covariate shift

The function `covariate_shift(severity, group)` implements a covariate shift that affects the scale of a particular feature dimension for one group only. We treat `severity`  $\gamma$  as a multiplicative scale factor applied to the standard deviation of that feature for the chosen group, with `severity`  $\in [1, 4]$ . Formally,

- if `group='A'`, we set `cov_scale_a = severity`, `cov_scale_b = 1.0`;
- if `group='B'`, we set `cov_scale_a = 1.0`, `cov_scale_b = severity`.

These parameters are passed to `generate_data`, which scales the feature standard deviation accordingly. The labels and group labels themselves are left unchanged.

This family of shifts isolates the effect of within-group feature distribution changes on fairness metrics, without directly altering label noise or group proportions. This mimics a setting where the target hospital's measurement or documentation practices differ for one demographic group (e.g., different laboratory equipment or measurement protocols), resulting in a change in the marginal distribution of a clinical feature for that group only.

### A.3 Group-specific label noise shift

Finally, the function `label_shift(severity)` modifies the label noise asymmetrically across groups. Here, `severity` (chosen in  $[0, 0.3]$ ) controls the probability with which positive labels for group A and, at half that rate, negative labels for group B are flipped. Concretely, we call

- `generate_data(flip_noise_a=severity)`,

which flips positive labels in group A with probability  $\beta$  and negative labels in group B with probability  $\beta/2$ . Features are generated before these flips, so the marginal  $X$  distribution is unchanged even though the relationship between observed labels and features is altered asymmetrically.

This corresponds to asymmetric diagnostic coding errors across groups in the target hospital (e.g., differential mis-documentation of follow-up need in electronic health records for one group).

Together, these three mechanisms allow us to separately study changes in group-specific base rates, changes in group-specific feature distributions, and changes in group-specific label noise, and to observe how standard fairness and calibration metrics respond when each dimension is varied in isolation.

## B Full Fairness and Calibration Metric Definitions

This appendix provides the formal definitions for the metrics summarized in Section 2.2.

We focus on three standard group fairness quantities implemented via `fairlearn.metrics(src/metrics.py)`:

- **Demographic parity difference (DP diff).** We measure the absolute difference in positive prediction rates between the two groups:

$$|\Pr(\hat{Y} = 1 \mid S = 0) - \Pr(\hat{Y} = 1 \mid S = 1)|.$$

A value of 0 corresponds to perfect demographic parity.

We compute this quantity using `fairlearn.metrics.demographic_parity_difference`. In clinical settings, a large DP difference means that one group is receiving more referrals than the other.

- **Equalized odds difference (EO diff).** We measure the worst-case discrepancy across groups in both true positive and false positive rates:

$$\max(|\text{TPR}_0 - \text{TPR}_1|, |\text{FPR}_0 - \text{FPR}_1|).$$

A value of 0 corresponds to perfect equalized odds.

We compute this quantity using `fairlearn.metrics.equalized_odds_difference`. In our setting specifically, large EO difference means that either missed follow-up (TPR gap) or unnecessary referrals (FPR gap) are distributed unevenly between groups.

- **True-positive-rate gap (TPR gap).** We separately report  $|\text{TPR}_0 - \text{TPR}_1|$ , the equal-opportunity violation. This is necessary for interpreting the threshold-adjustment experiment: a smaller TPR gap can coincide with a larger EO difference if the false-positive-rate gap grows. In the medical follow-up context, a large TPR gap implies that one group is systematically less likely to be referred for necessary follow-up, which is a direct equity-of-care concern.

In addition to group fairness, we track probability calibration at both the global and group levels. We compute the expected calibration error (ECE) following a standard binned estimator. Given predicted probabilities  $p_i$  and labels  $y_i$ , we partition the interval  $[0, 1]$  into  $B$  bins, and for each bin  $b$  compute the average predicted confidence and empirical accuracy. The ECE is then

$$\text{ECE} = \sum_{b=1}^B \frac{n_b}{N} |\text{acc}(b) - \text{conf}(b)|,$$

where  $n_b$  is the number of points in bin  $b$ ,  $\text{acc}(b)$  is the average label, and  $\text{conf}(b)$  is the average predicted probability in that bin.

To capture group-wise calibration, we compute ECE separately for each group and report the ECE gap, defined as the difference between the worst- and best-calibrated group:

$$\text{ECE Gap} = \max_a \text{ECE}_a - \min_a \text{ECE}_a.$$

This is implemented in `src/metrics.py` as `calculate_group_ece_metrics`, which returns both the per-group ECEs and their gap. A large ECE gap indicates that for some groups, predicted follow-up risks are systematically over- or under-confident, which may interact with thresholding policies in opaque ways. We also report balanced accuracy to expose utility changes that may accompany an apparent fairness improvement.

## C Full Mitigation Strategy Details

This appendix provides the complete implementation details for the three interventions summarized in Section 2.3.

To evaluate whether fairness distortion can be detected or corrected, we implement three post-hoc and training-time interventions:

- **KDE Importance Reweighting.** We estimate the density ratio  $w(x) = \hat{p}_{\text{target}}(x)/\hat{p}_{\text{train}}(x)$  using an independent unlabeled target adaptation sample and kernel density estimation (`src/reweighting.py`). The features are jointly standardized, weights are clipped to  $[0.1, 10]$ , and the logistic-regression loss is reweighted. This approach is most directly justified under covariate shift.
- **Post-hoc Threshold Adjustment (Equal Opportunity).** Using an independent labeled calibration set, we take the larger group TPR at threshold 0.5 as the target and choose a deterministic threshold for each group on a grid with spacing 0.01 (`src/adjust_threshold.py`). This attempts to raise the lower-TPR group without retraining the classifier. It does not enforce full equalized odds and can increase the FPR gap. This is analogous to a hospital administrator adjusting group-specific referral thresholds to meet an equal-access policy, while monitoring unintended consequences on other metrics.
- **Target-Distribution Retraining.** As a supplementary reference, we retrain the same model family on an independent sample from the shifted distribution and evaluate it on the shifted test set. This measures what target-distribution ERM can attain relative to a frozen source model; it is not a universal fairness optimum or a guaranteed improvement, since the same logistic model class and ERM objective may still be misspecified under the shifted distribution.

## D Full Implementation Details

This appendix provides the complete experimental implementation details complementing Section 3.1.

- Sample size: source training, shifted evaluation, and target-retraining datasets each use  $N = 5000$ . KDE adaptation and threshold calibration each use independent samples of size 1000.
- Classifier: Logistic regression with `max_iter=1000` and default L2 regularization (`C=1.0`).
- Fairness and calibration metrics: fairness metrics are computed via `fairlearn.metrics`; ECE uses 10 equal-width bins.
- Repetitions: all results are reported for a fixed random seed (42); diagnostics (ESS and thresholds) are logged for transparency.

## E Supplementary Results Figures

This appendix provides supporting visualizations for the results summarized in Section 4.

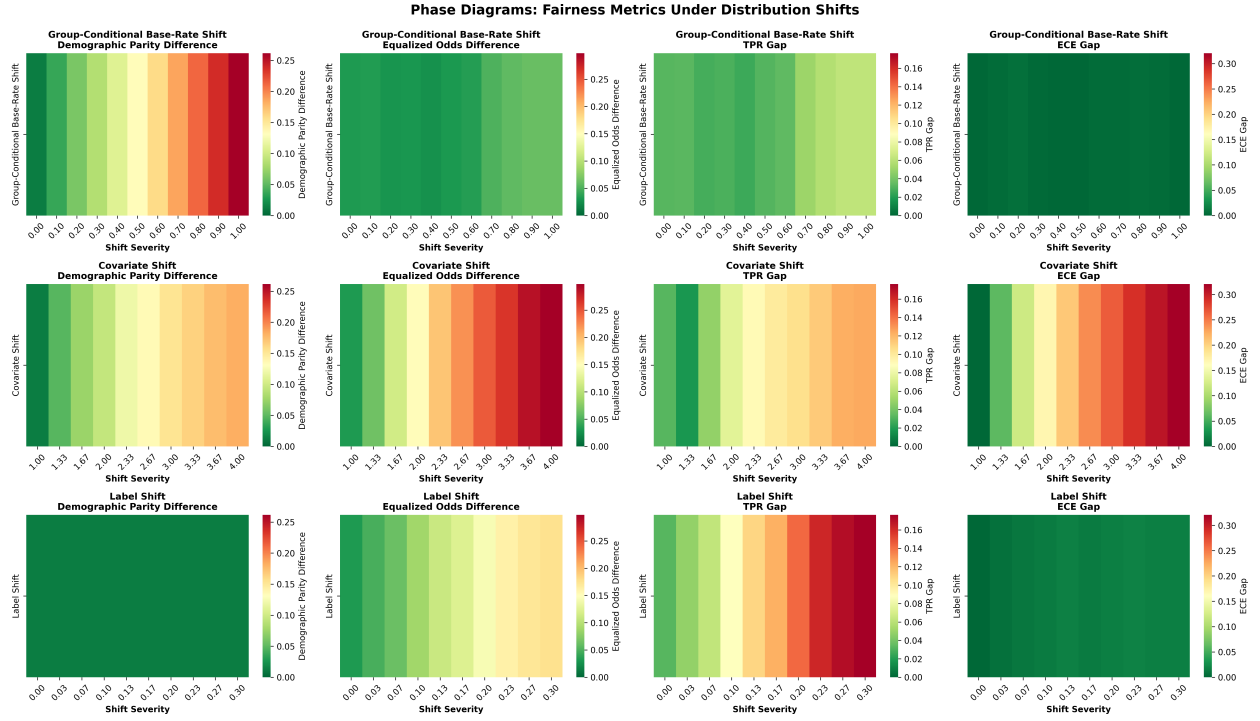


Figure 1: Metrics vs. severity for different shifts. For group-conditioned base-rate shift, DP increases substantially while TPR gap remains nearly flat. For group-specific covariate shift, all metrics worsen, but at different rates; ECE gap reaches 0.32. For asymmetric label noise. DP remains artificially flat while TPR gap rises sharply.

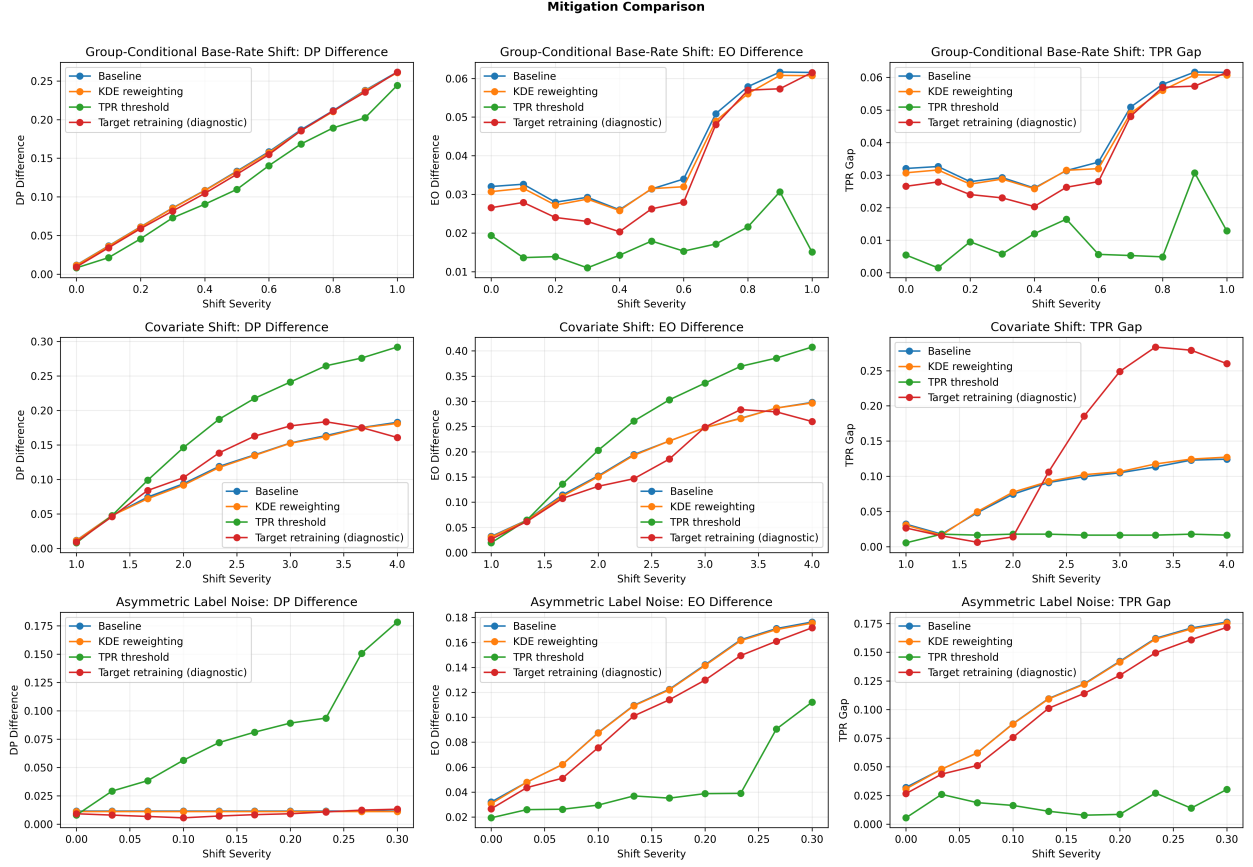


Figure 2: Comparison of mitigation strategies (KDE reweighting, threshold adjustment, and target retraining) relative to the frozen baseline under different shifts. Threshold adjustment reduces TPR gap at the cost of increasing EO.